



Infections per student in a hybrid-flexible vs. traditional class setting

UNIVERSITY OF ZURICH

BIO369 AGENT-BASED MODELLING, FS 2023

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Background

Traditional setting (brick-and-mortar)



- Classes are on-site only
- Increases consumption of energy, heating, and building space (Roy et al., 2008)
- May not be accessible for students with daytime jobs or health conditions (Beatty, 2019)
- May cause students to miss lecture content due to sickness, or impel students to attend lectures despite being sick
- Facilitates spread of disease (Hekmati et al., 2022)

HyFlex (= hybrid-flexible)



- Students choose between attending a lecture in person vs. remotely (Beatty 2019)
- Reduces energy usage and CO₂ emissions (Roy et al., 2008)
- Makes higher education more inclusive (Beatty, 2019)
- Provides additional learning opportunities through podcasts (Beatty, 2019)
- May help mitigate spread of infectious diseases (Hekmati et al., 2022)

Aim

Research Questions

- How much can a HyFlex class setting lower the average number of infections per student, compared to a traditional setting?
- How many classes does the average student miss due to being sick from getting infected in a traditional setting?
- How much space can the university save with a HyFlex class setting?

Methods

Flow

- For 100 iterations
 - For traditional and hyflex setting
 - For each number of rows
 - For 14 weeks
 - Simulate transmission
 - Accumulate `n_infections` and `n_missed_classes`
 - Summarize `n_infections` and `n_missed_classes` over all weeks

General assumptions

- Square classrooms, 300 students (Assumed)
- High season from September to April (Pappas, 2018). Simulation: 14 weeks
- Immunity for 3 weeks after recovery (Eggo et al., 2016)
- R_0 in a traditional class setting: 3 (Hekmati et al., 2022)
- Transmission threshold: 2.2 m (Khosronejad et al., 2021)
- Average number of neighbors within transmission threshold: ~ 11
- $\text{Beta} = 3 / 11 \approx 0.27 \rightarrow 0.2$

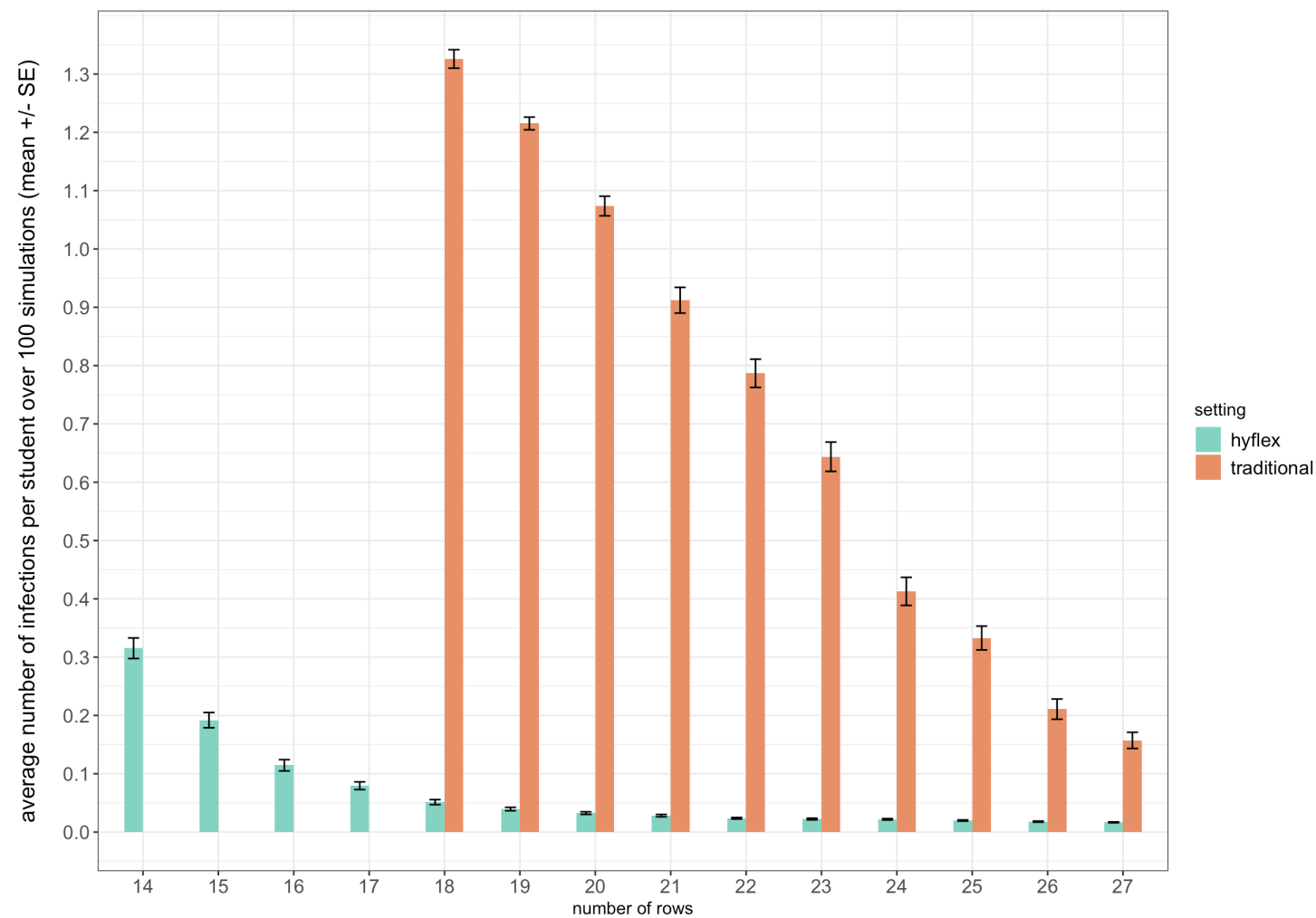
Specific assumptions

Traditional
Initial infection probability: 0.01 (Hekmati et al., 2022) → 3 patients zero
On-site only
If sick: $p = 0.2$ to attend the lecture anyway, $p = 0.8$ to miss the class

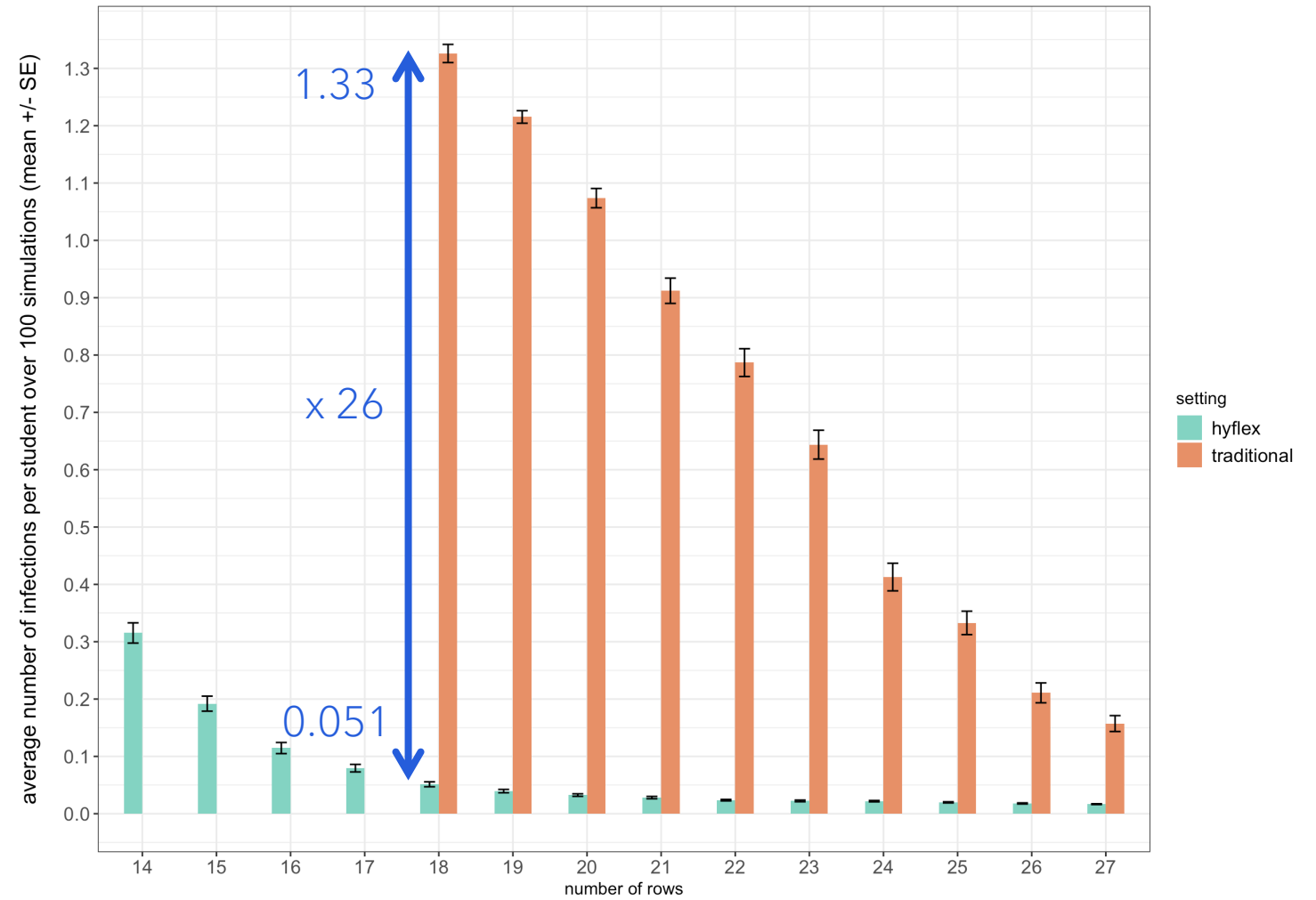
HyFlex
2 patients zero
1/3 always on-site 1/3 always online 1/3 hybrid (50/50)
No missed classes (podcasts)

Results

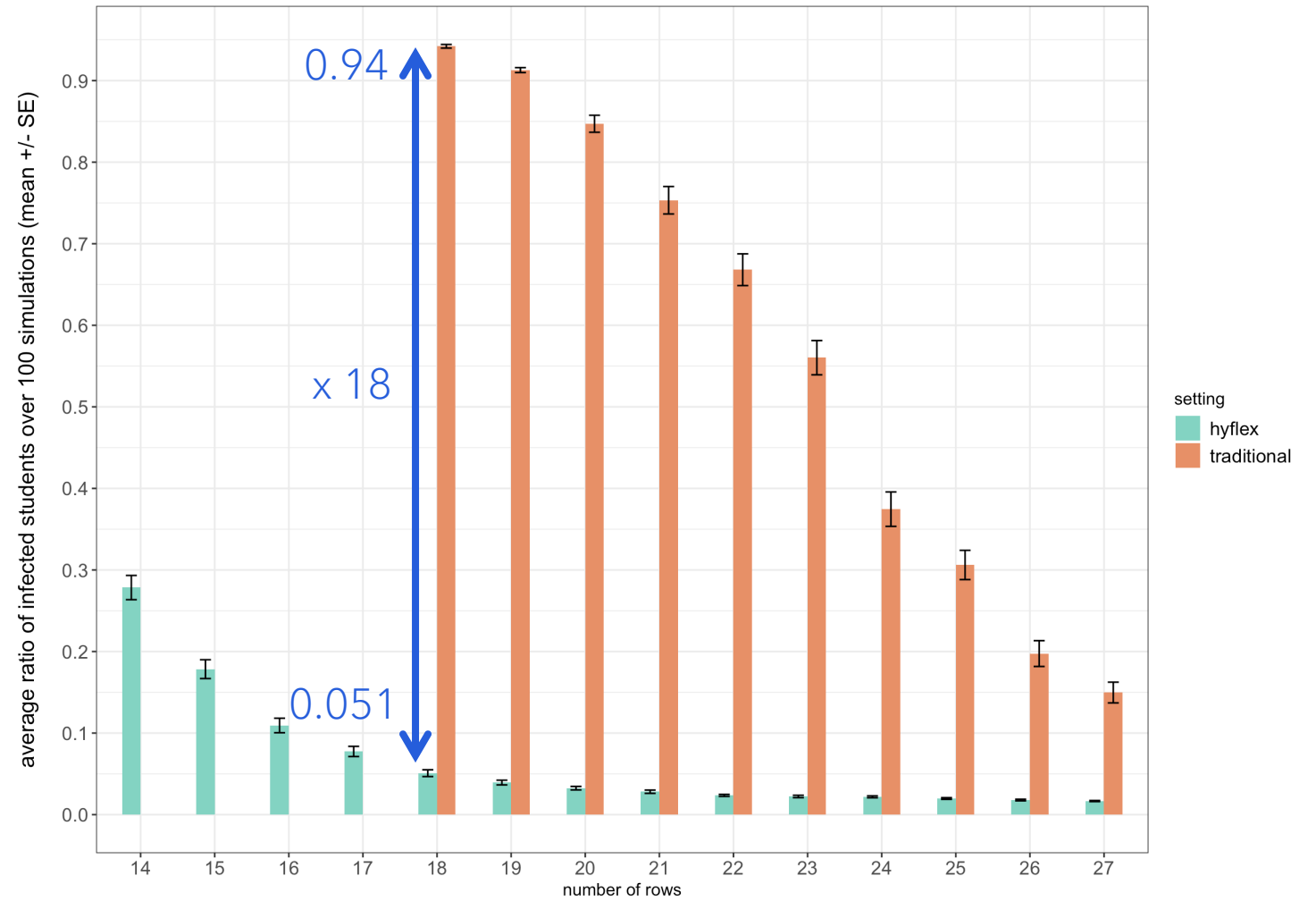
Infections per student



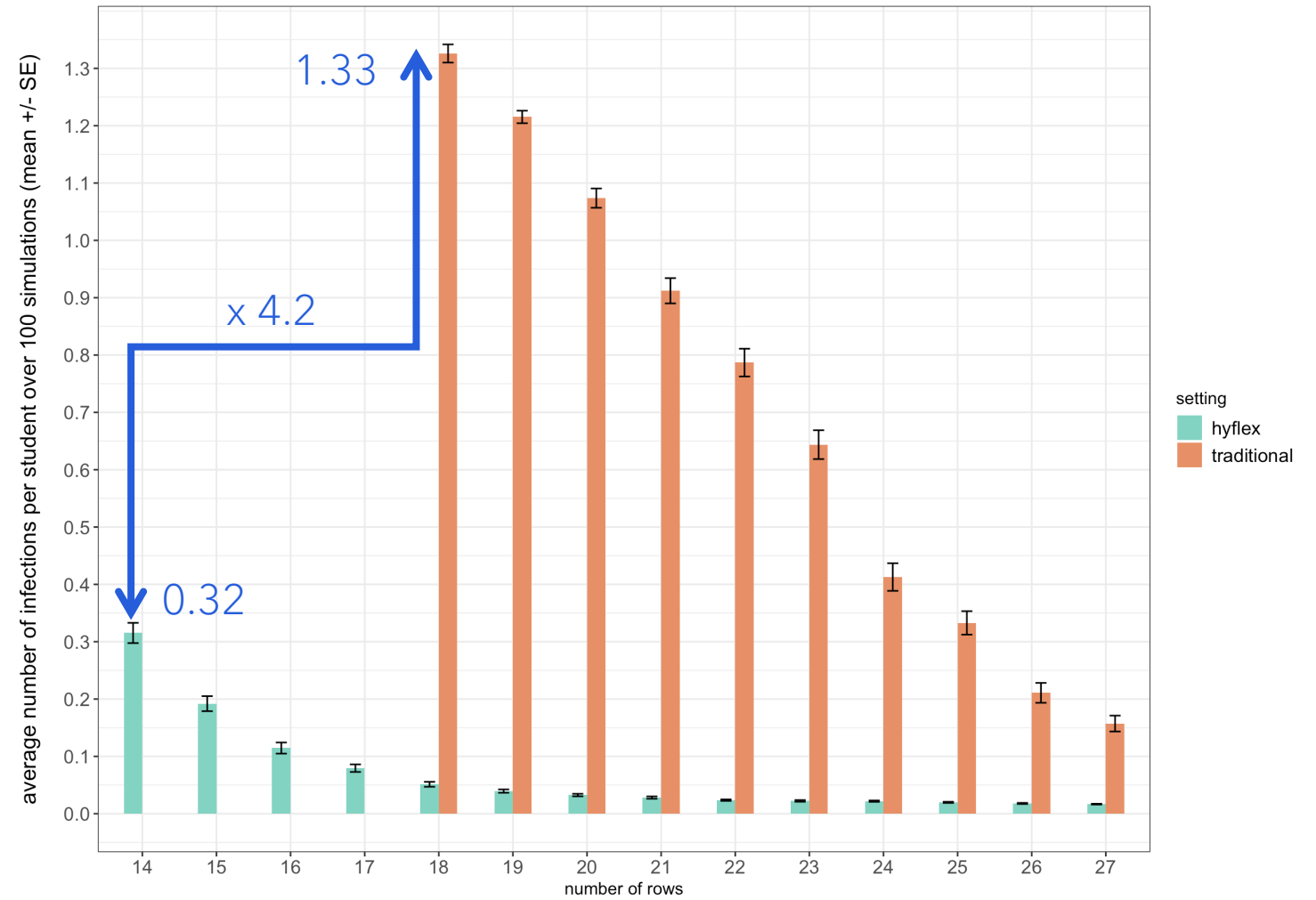
Infections per student



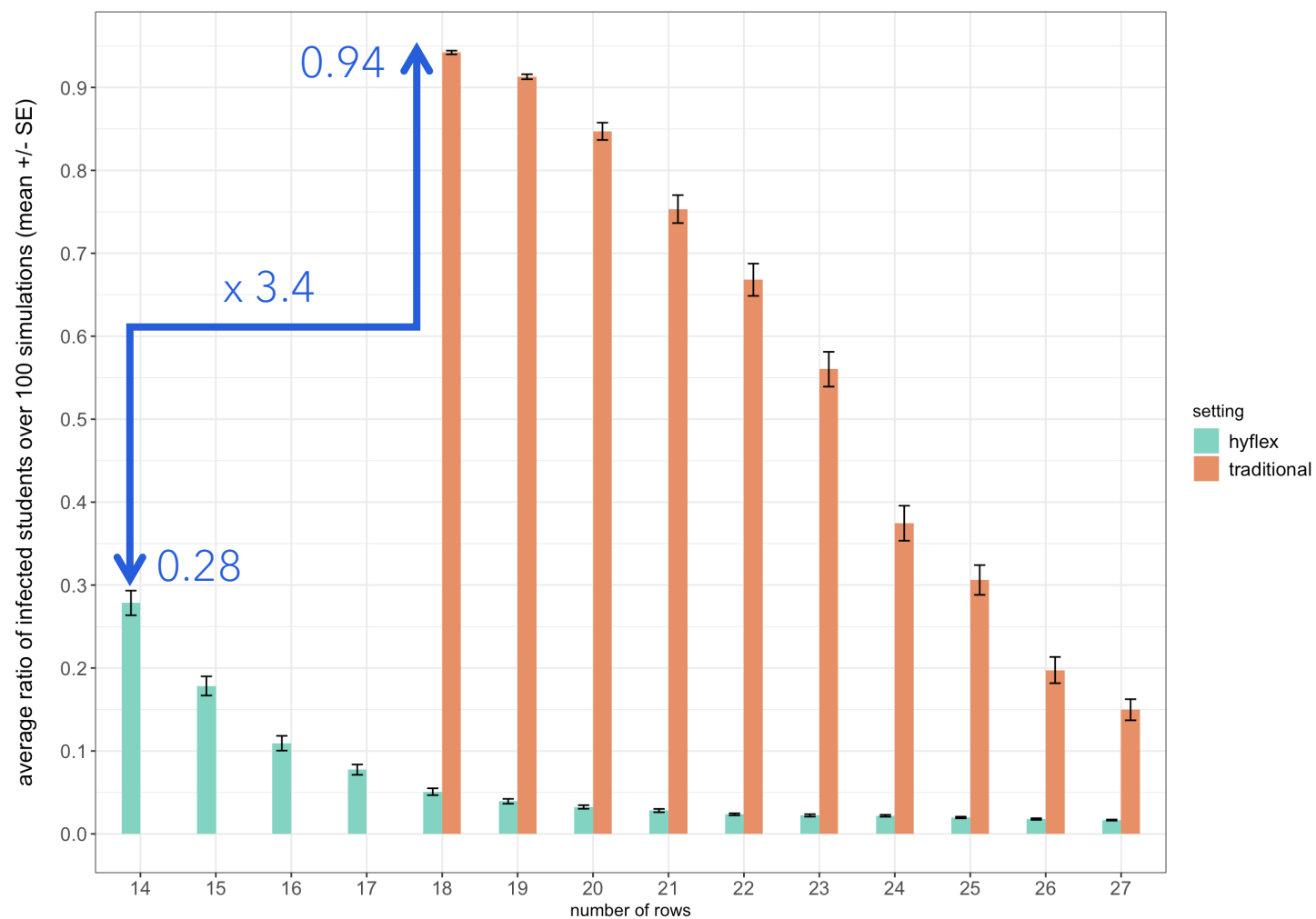
Ratio of infected students



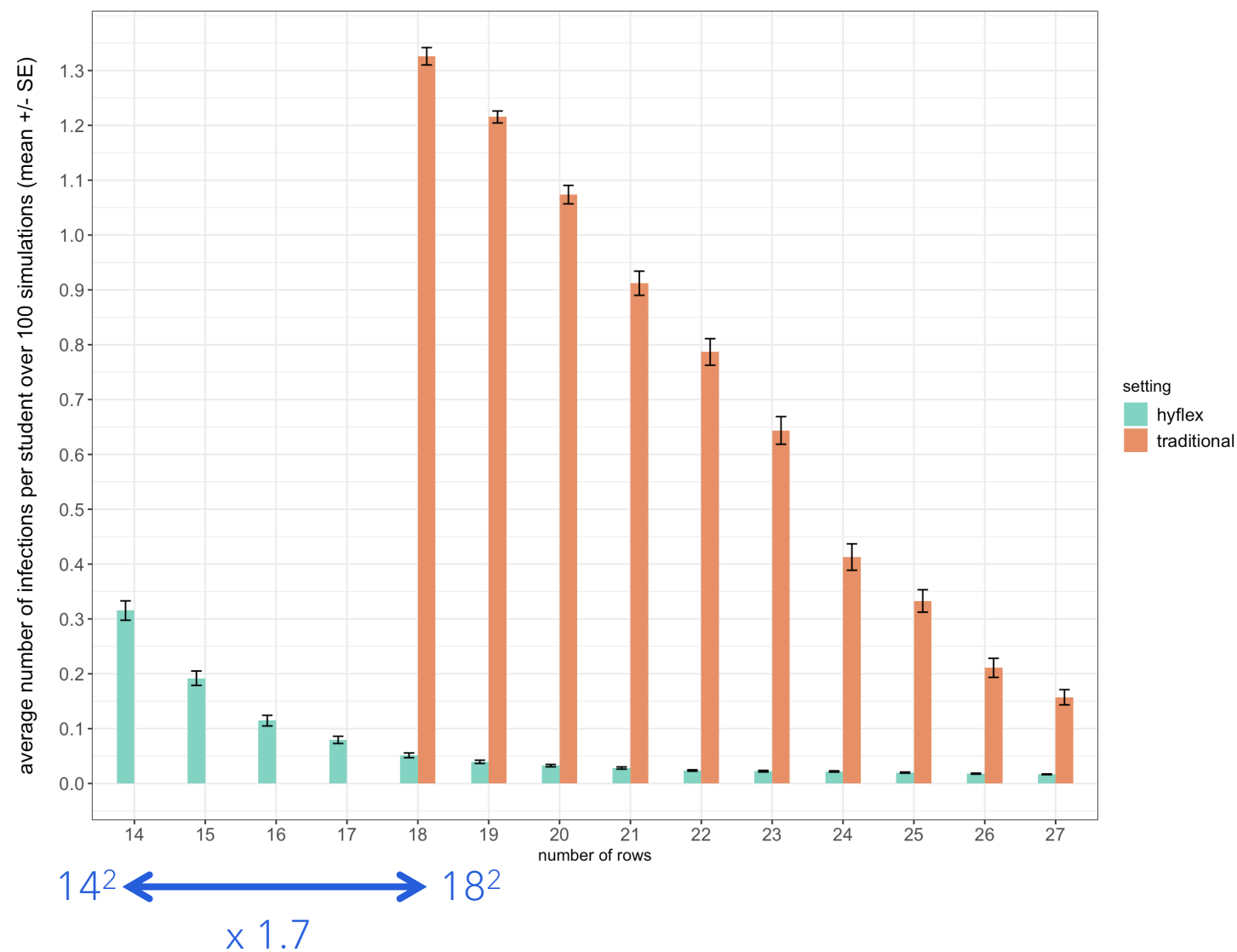
Infections per student



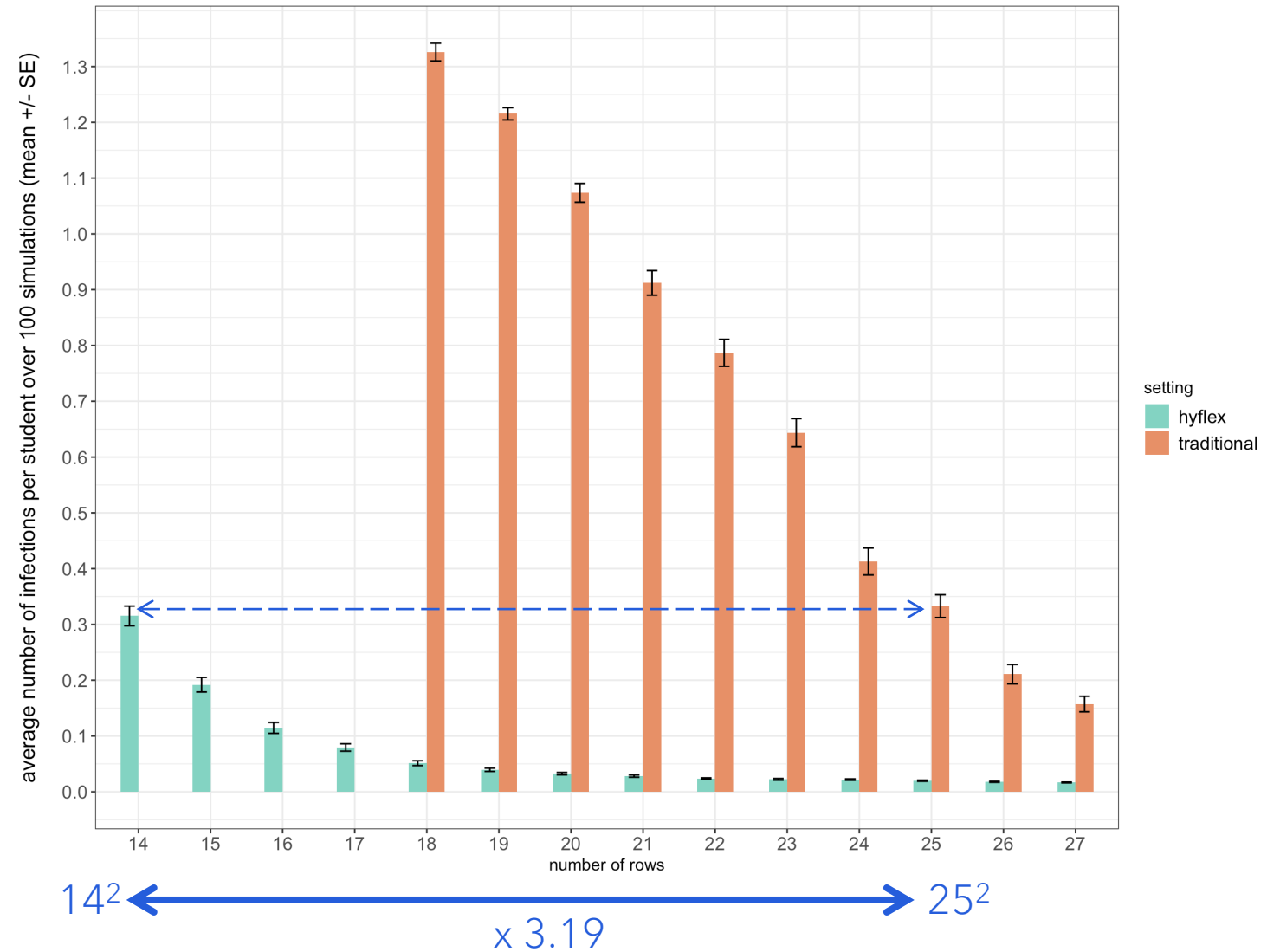
Ratio of infected students



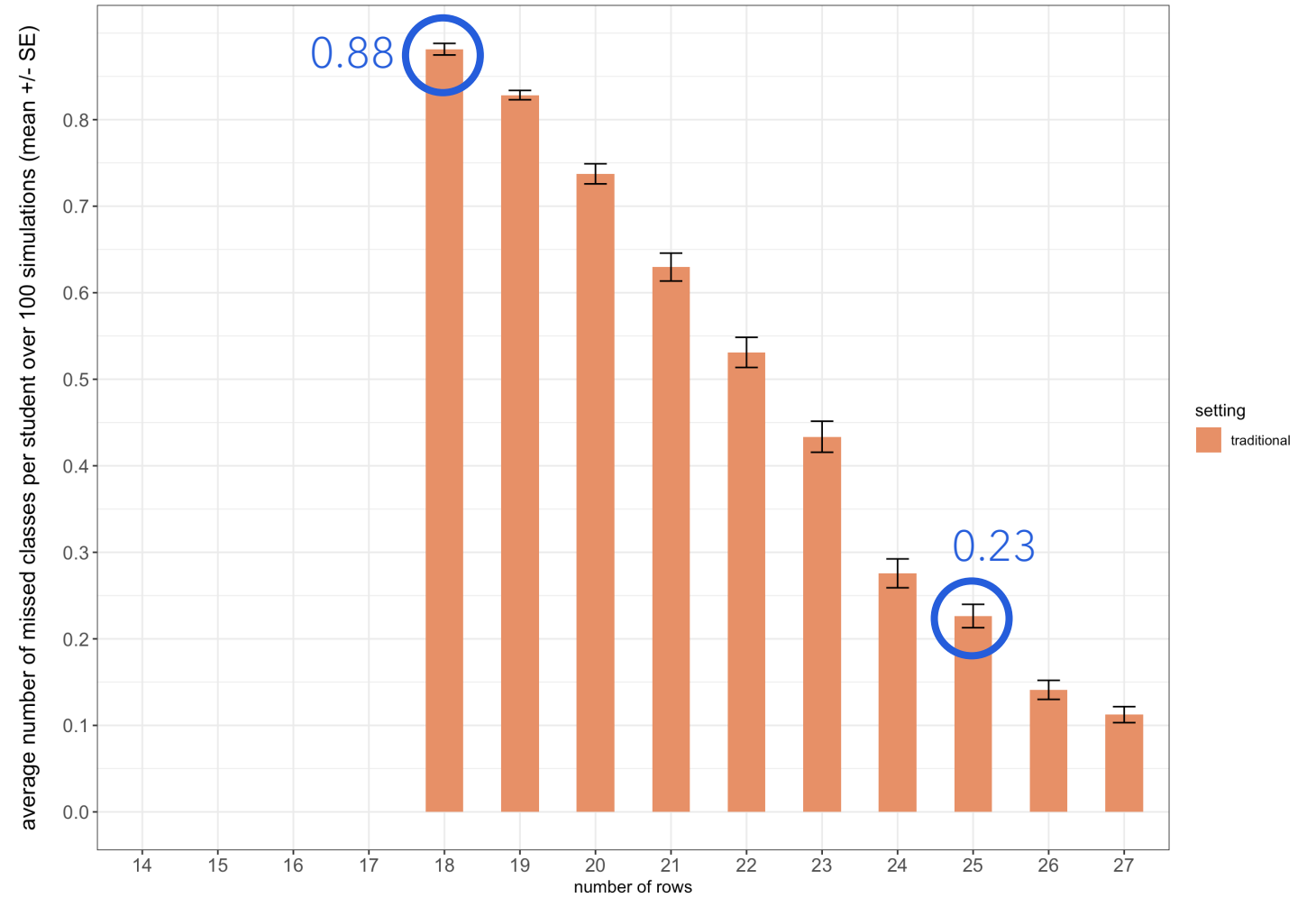
Infections per student



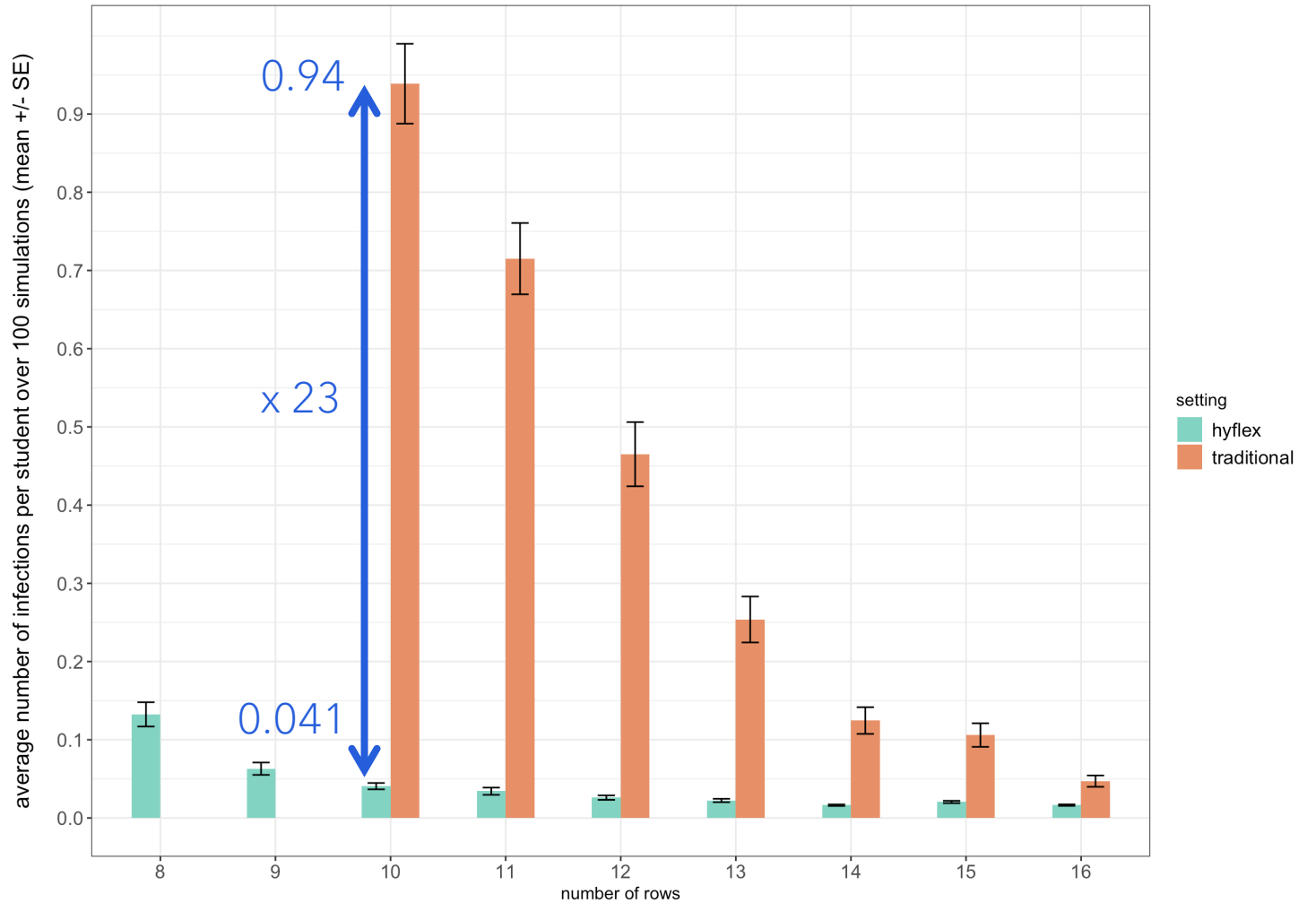
Infections per student



Missed classes



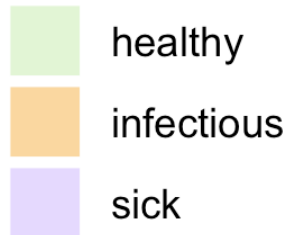
Infections per student (n = 90)



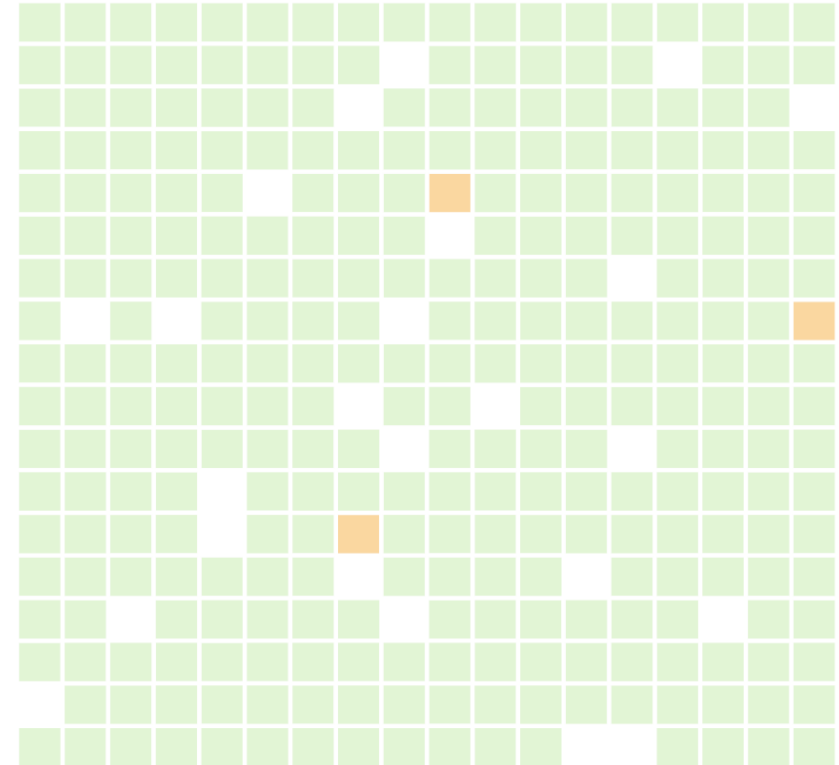
Traditional

18 x 18 rows

status

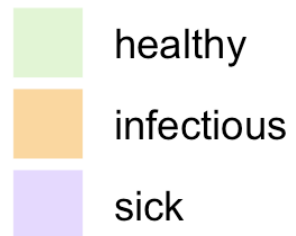


week 01



Traditional 18 x 18 rows

status

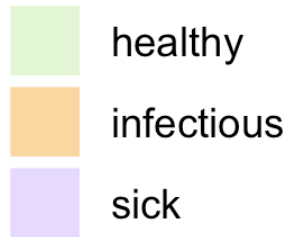


week 01

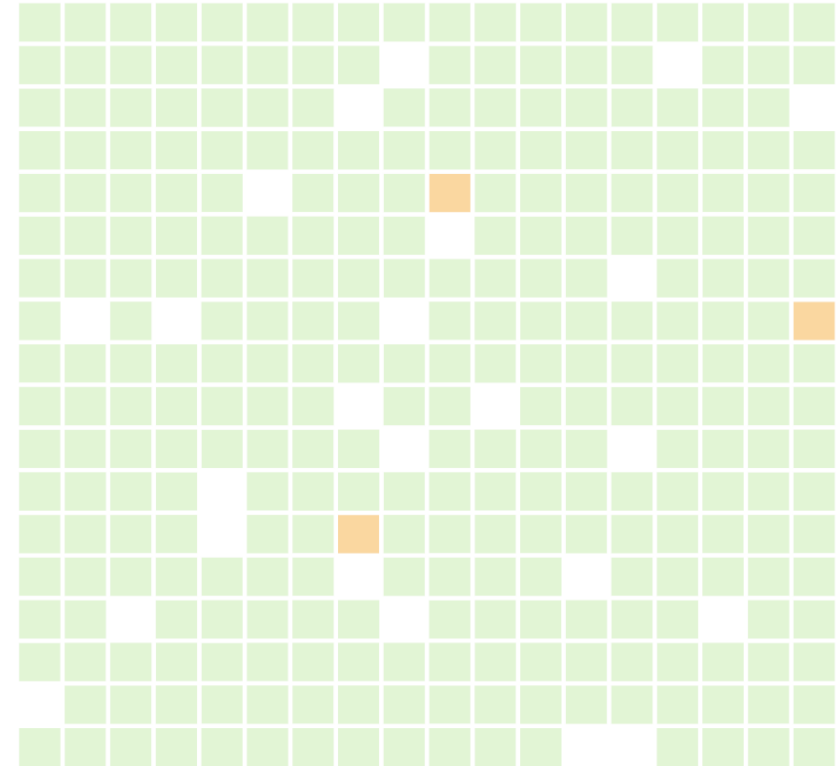
Traditional

18 x 18 rows

status



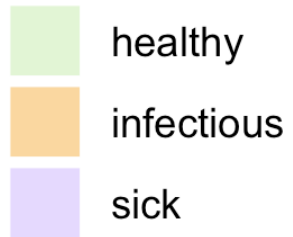
week 01



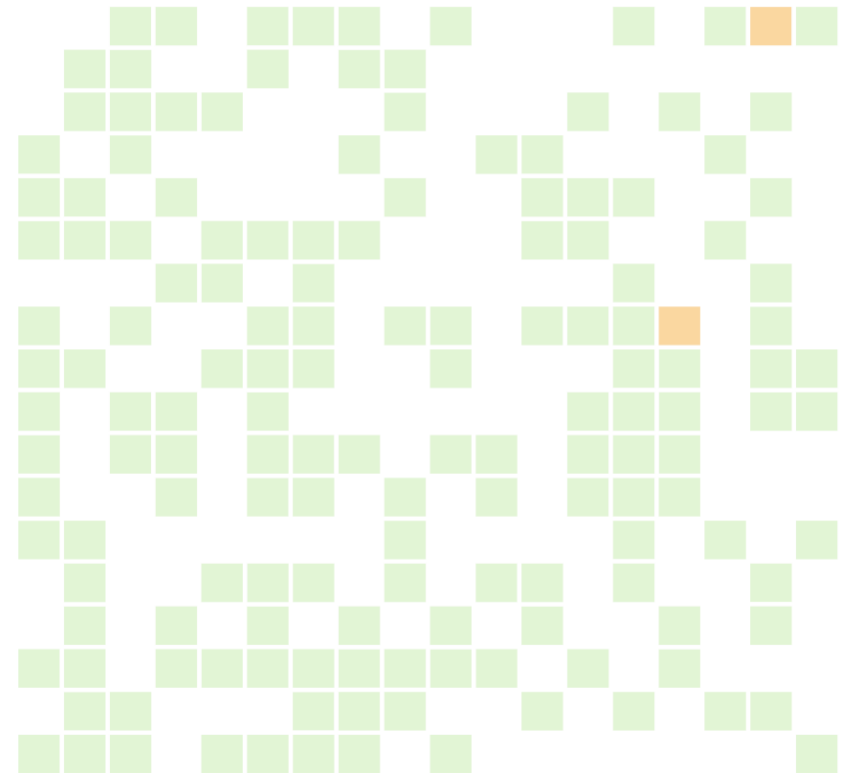
HyFlex

18 x 18 rows

status



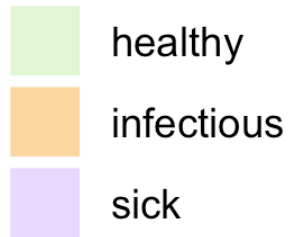
week 01



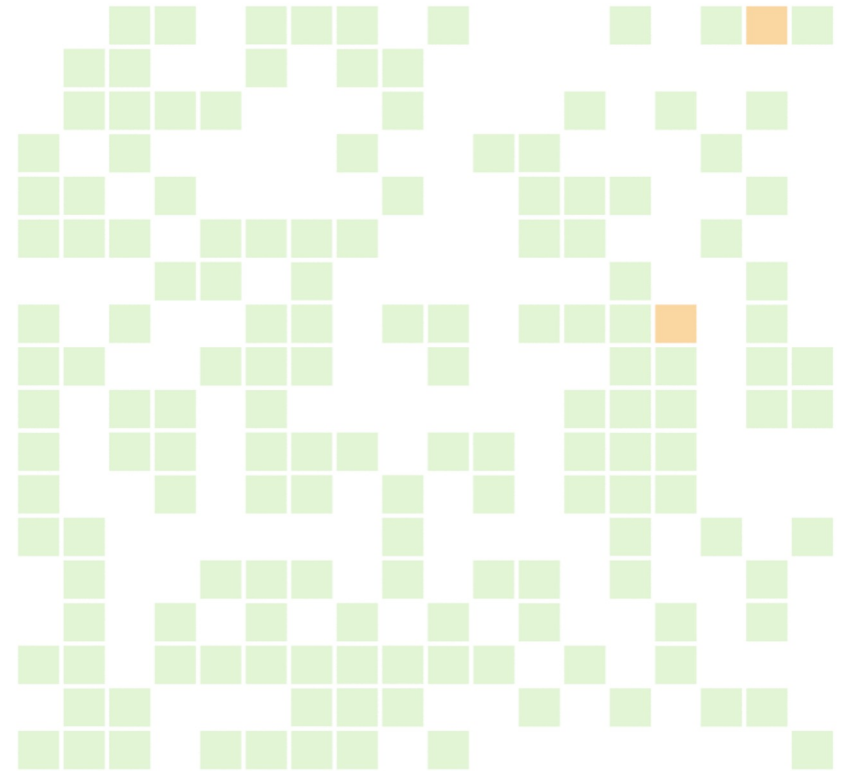
HyFlex

18 x 18 rows

status



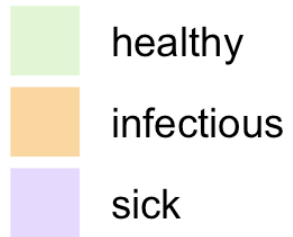
week 01



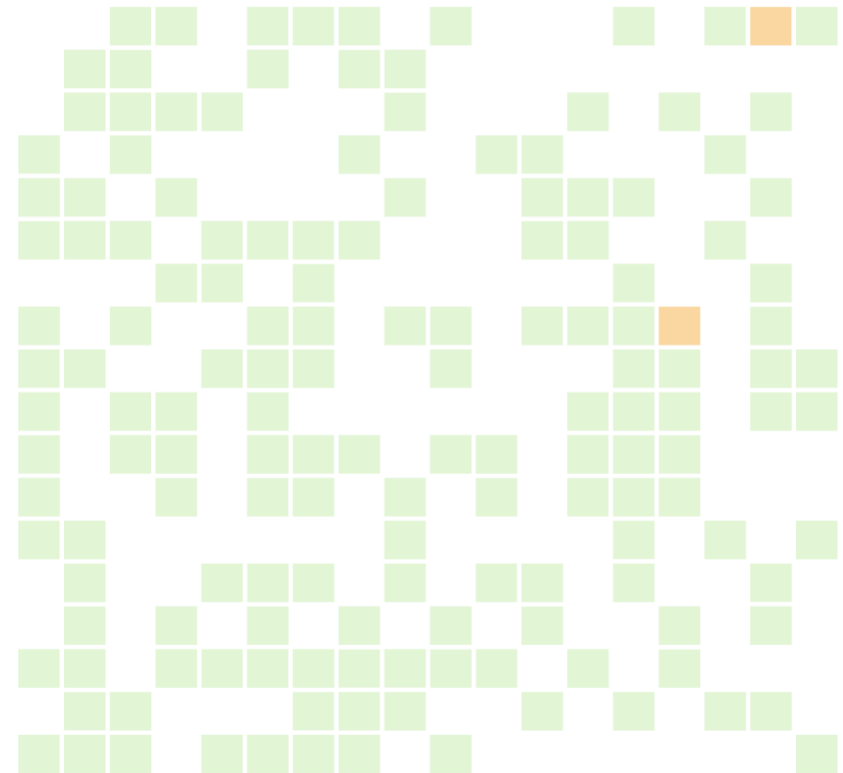
HyFlex

18 x 18 rows

status



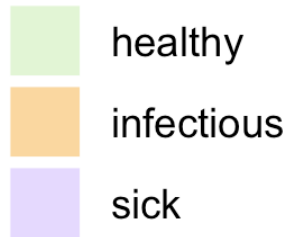
week 01



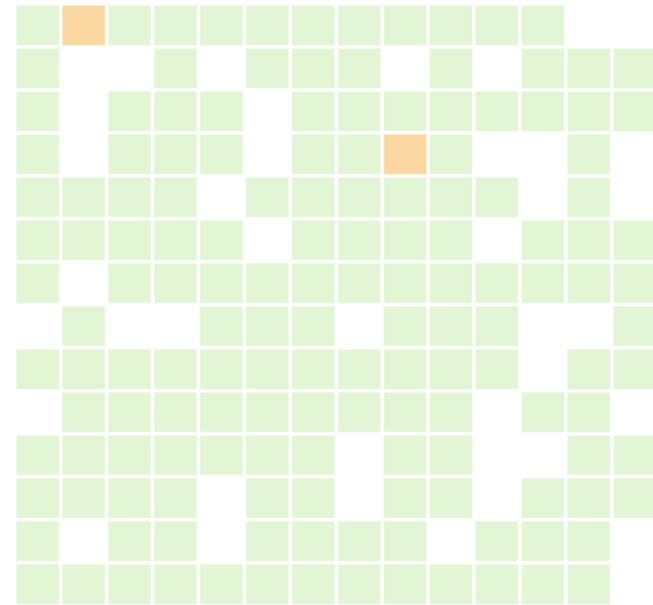
HyFlex

14 x 14 rows

status



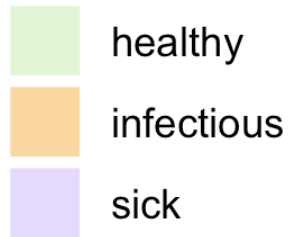
week 01



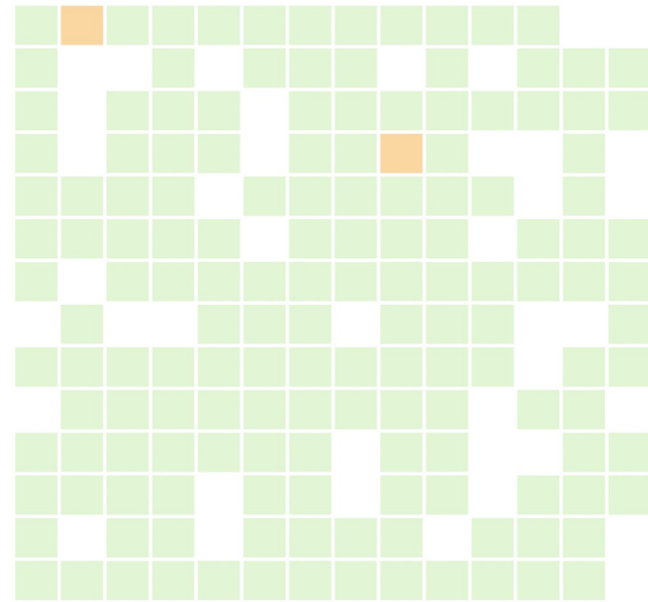
HyFlex

14 x 14 rows

status



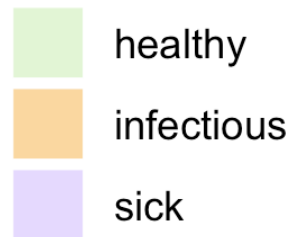
week 01



HyFlex

14 x 14 rows

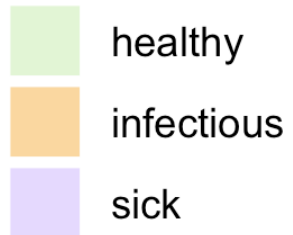
status



week 01

Traditional 25 x 25 rows

status

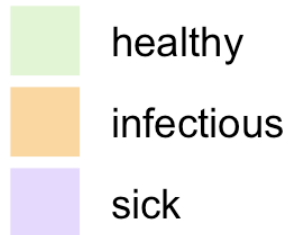


week 01

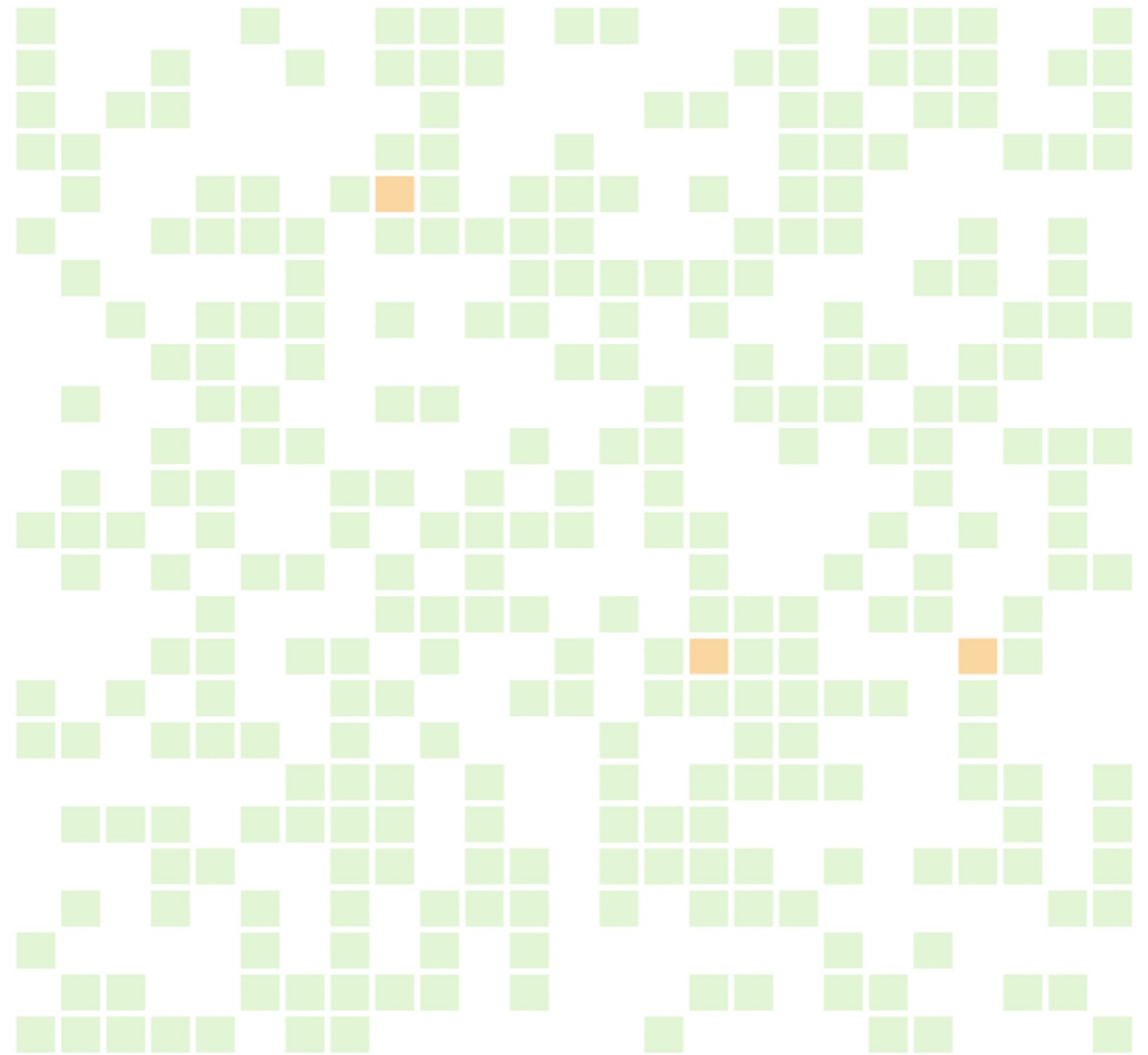


Traditional 25 x 25 rows

status

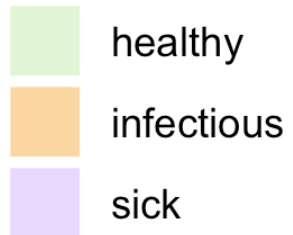


week 01



Traditional 25 x 25 rows

status



week 01



Result

In a classroom with 18 rows (current state assumed), the number of infections per student is 26 times less (3.8%) in the hyflex setting compared to the traditional setting (0.051 vs. 1.33).

In the smallest classroom possible for the HyFlex setting (14 rows), the number of infections would be 4.2 times less (24%) than in the smallest possible classroom for the traditional setting (18 rows), 0.32 vs .1.33.

In the hyflex setting, the minimum number of rows is 14, while for the traditional setting it is 18. That means the minimum requirement of space for the hyflex setting is 1.65 times less (60%).

For the traditional setting to reach a similar infection rate as the highest infection rate in the hyflex setting (0.32 infections per student in a classroom with 14 rows), it would require 25 rows. This means the hyflex setting requires 3.19 times less space (31%) than the traditional setting.

With the minimum number of rows in the traditional setting (18), the average student misses 0.88 classes. With 25 rows, they miss 0.23 classes.

Conclusions

Conclusions

- HyFlex can drastically reduce the number of infections per student compared to a traditional setting
- HyFlex allows for smaller classroom sizes compared to traditional classes, thus potentially cutting the university's costs for energy, heating, and maintenance
- In the traditional setting, some classes are missed due to sickness

Limitations

- Does not take into account
 - Duration of classes (Wang et al., 2021)
 - β and thresholds depending on behaviour (masks, coughing, etc.) (Arumuru et al., 2020)
 - dynamics of aerosols and ventilation (Wang et al., 2021) (Hekmati et al., 2022)
 - different betas and transmission thresholds of different viruses
 - transmission in public transport
 - etc.

References

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Thank you for listening.